

Fig. 1. Biasing circuit

Explanation:

In general, circuits require their common-source and cascode transistors to be biased with a certain gate-voltage. Concentrating on the NMOS (bottom) side of the circuit, you can see 3 transistors: M_{n1} , M_{n2} and M_{n3} . M_{n1} is intended to bias the gate of common-source amplifiers (or current-sources) and the combination of M_{n2} and M_{n3} is intended to bias the gate of cascode transistors. Generally speaking, M_{n1} and M_{n2} are sized equally. As they have the same drain-current, they then also will have the same V_{gs} . A similar reasoning also holds for the PMOS-side with M_{p1} , M_{p2} and M_{p3} .

Simulation 1:

Please simulate the circuit of fig. 1 and use for all transistors (**4 points**):

- ◆ $W = 1.00\mu$ (**Please use this for all your simulations in this course!**)
- ◆ $L = 0.18\mu$ (**Please use this for all your simulations in this course!**)
- ◆ M-factors to be decided by you, but keep:
- ◆ $M_{p1} = M_{p2} = M_{p3} = M_p$
- ◆ $M_{n1} = M_{n2} = M_{n3} = M_n$

Use $I_{b,main} = 200 \mu A$.

Choose M_n and M_p such that

- ◆ $V_{gtn1} = V_{gtn2} \sim 100 \text{ mV (approx.)}$
- ◆ $V_{gtp1} = V_{gtp2} \sim 100 \text{ mV (approx.)}$

Determine (sweep) $I_{b,ncas}$ such that M_{n1} is on the edge of saturation, with $\sim 50\text{mV}$ margin. Also determine (sweep) $I_{b,pcas}$ such that M_{p1} is on the edge of saturation, with $\sim 50\text{mV}$ margin. Theoretically, to achieve this $I_{b,ncas}$ and $I_{b,pcas}$ need to be $4I_{b,main}$, but in practice it will be somewhat different. Please figure out what the correct ratios are and explain why (**1 point**). Note that the optimal value for the ratios for n and p can be different.

After determining the correct ratio, use the following expression for $I_{b,ncas}$ and $I_{b,pcas}$:

- ◆ $I_{b,ncas} = \text{ratio}_n I_{b,main}$ and
- ◆ $I_{b,pcas} = \text{ratio}_p I_{b,main}$.

Keep this set-up for **all your following simulations**, such that you can vary $I_{b,main}$ and both $I_{b,ncas}$ and $I_{b,pcas}$ will follow automatically. Simulate what happens if you reduce the currents by 2x and by 4x (**1 point**).

Show a clear circuit-diagram with all node voltages and branch currents.

Simulation 2:

Use the circuit of fig. 1 to build the circuit of fig. 2, by just adding transistors Mn4, Mn5, Mp4 and Mp5. Use the same size for these transistors as in the biasing circuit. Use voltage sources for V_{op} and V_{on} . Sweep both V_{op} and V_{on} individually and measure the current in each branch. Make a plot of $I(V_{op})$ versus V_{op} and $I(V_{on})$ versus V_{on} (1 point). This can be done in 1 single plot, with V_{op} and V_{on} varying from 0 to 1.8 V. Determine (roughly) for both branches what the minimum voltage is across the 2 transistors (so either Mn4 and Mn5 or Mp4 and Mp5) to keep the devices saturated. Use in this simulation the exact same size and current-levels as in simulation 1 (1 point).

Also plot in the same figure the output impedance of each branch by taking 1 over the derivative of the current versus voltage curve (2 points).

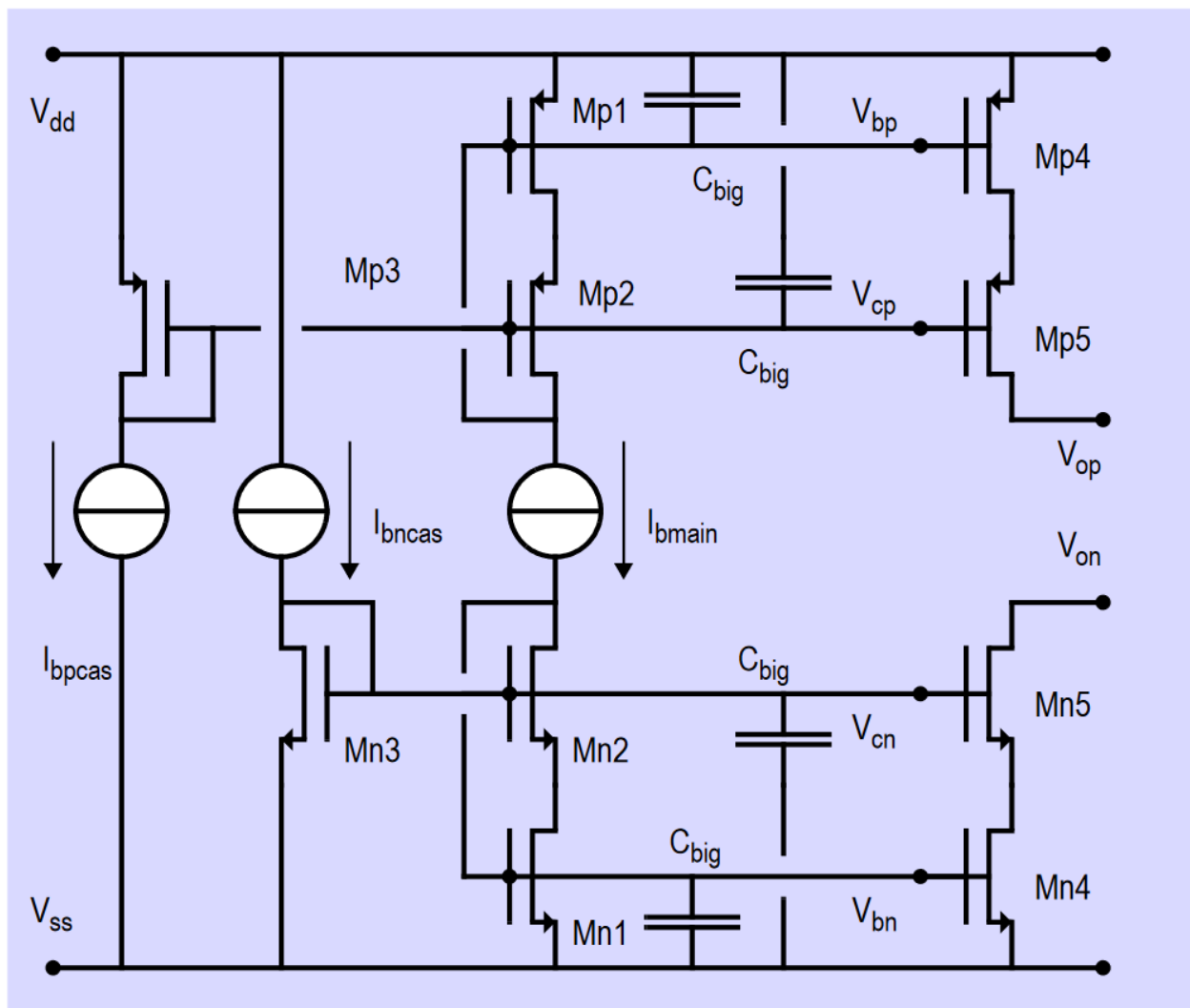


Fig. 2. P- and N- Current-sources

Remarks:

- When reusing this block for follow-up simulations, please always use **big capacitors (C_{big})** to decouple the biasing node and prevent the circuits we are biasing from changing the voltages on the bias nodes.
- For the sake of simulation, very large capacitors (like 1 F) are allowed, even if those are not practical value.
- Please make sure that
 - ♦ $V_{\text{gtn}1} = V_{\text{gtn}2} \sim 100 \text{ mV (approx.)}$
 - ♦ $V_{\text{gtp}1} = V_{\text{gtp}2} \sim 100 \text{ mV (approx.)}$
- Although M-factors are a number of fingers and as such they are **integer** numbers, **you will be allowed to use non-integer (fractional) numbers**. The above recommendation of the V_{gt} 's $\sim 100\text{mV}$ cannot be met exactly in practice and therefore it is also not a requirement in this homework to meet this spec exactly. This exercise is also not an exact science and you don't have to optimize your transistor sizes to get the last fraction of a mV for either a V_{gt} or for the margin you have on $V_{\text{ds}} - V_{\text{dsat}}$.
- Please connect the **backgates (bulk contacts)** of all nmos devices to V_{ss} (0 V) and the backgates of all pmos devices to V_{dd} (1.8 V).
- Note that when re-using this block for other parts of this project or for the project of EE4525, you may want to use different values of V_{gt} . If this homework is done correctly, using other V_{gt} 's can simply be obtained by using a different value for $I_{\text{b,main}}$, while maintaining the correct values for ratio_n and ratio_p .
- Please submit the report only.